Problem 6.16

Journal entries — perpetual and periodic inventory systems

Non-GST version

Assume that Merryland's Markets had an inventory balance of $32 570 at the close of the last accounting period. The following sales and purchase transactions are for the current period.

1. Purchased goods on account for $27 190.
2. Returned part of the above purchase that had an original purchase price of $1590.
3. Paid for the balance of the purchase in time to receive a discount of 2% of the purchase price.
4. Sold goods costing $24 900 for $49 820. Cash of $23 000 was received, with the balance due on account.
5. Goods sold on credit for $2026 (cost $1010) were returned.

Required

(a) In two columns, prepare general journal entries (ignoring GST) assuming:
   i. a periodic inventory system is used
   ii. a perpetual inventory system is used.

(b) Suppose that a physical count of the inventory at the end of the current period shows inventory of $30 000 to be on hand. Present the entries (if any) required under each inventory system to adjust for any discrepancy.

(c) Comment on which system would best disclose any discrepancy.

(LO3 and LO4)

(a)

MERRYLAND'S MARKETS

	Periodic			Perpetual		
1.	Purchases	27 190		Inventory	27 190	
	Accounts Payable		27 190	Accounts Payable		27 190
2.	Accounts Payable	1 590		Accounts Payable	1 590	
	Purchases Returns & Allows.		1 590	Inventory		1 590
3.	Accounts Payable	25 600		Accounts Payable	25 600	
	Discount Received		512	Discount Received		512
	Cash at Bank		25 088	Cash at Bank		25 088
4.	Cash	23 000		Cash	23 000	
	Accounts Receivable	26 820		Accounts Receivable	26 820	
	Sales		49 820	Sales		49 820
				Cost of Sales	24 900	
				Inventory		24 900
5.	Sales Returns & Allows	2 020		Sales Returns & Allows	2 020	
	Accounts Receivable		2 020	Accounts Receivable		2 020
				Inventory	1 010	
				Cost of Sales		1 010

(b)

Physical count at period end $30 000	Inventory balance Perpetual System $30 000
No entry required under periodic system. Inventory loss included in calculation of Cost of Sales.	$32 570 (balance) + $27 190 (purchase) – $1 590 (return) – $24 900 (sale) + $1 010 (sale return) = $34 280
	Inventory Shortage Expense 4 280 Inventory 4 280
	Inventory loss $43 280 – $30 000 = $4280

(c) The perpetual inventory system is the only system which can identify inventory discrepancies. This system maintains an accurate record of how many inventory units should be on hand at any one time. The system can then identify any discrepancy between recorded inventory balances and the stocktake or physical count of inventory numbers.

In this case the inventory shortage amount of $4280 is significant and the owner of the business needs to investigate the cause. Is it theft? Is the inventory being discarded because it has perished? Is it a combination of factors? What can be done about it?

Exercise 13.5

Inventory cost methods — perpetual inventory system

Non-GST version

The following information relates to the inventory of the Denki Ltd during the month of August:

August 1	Beginning Inventory	60 units @ $10
12	Purchased	90 @ $12
15	Sold	80 units
21	Purchased	120 @ $13
22	Sold	100 units
28	Sold	65 units

Denki Ltd uses the **perpetual inventory method** of accounting for inventory. Ignore GST.

Required

Determine the cost of the ending inventory. Note a physical stocktake verified inventory records reflected ending stock levels. The manager of Denki Ltd is evaluating inventory valuation methods and is seeking your assistance to calculate the value of closing inventory and cost of goods sold for each of the following methods:

(a) FIFO
(b) LIFO
(c) Moving average; note, round unit cost to the nearest cent.
(d) Specific identification – records identified the ending inventory consisted of 5 units from beginning inventory, 12 units from the purchase on 12 August and the remaining from the purchase on 21 August.
(LO3)

Suggested Solution:

(a)

(a). FIFO		Purchases			Sales			Balance		
Date	Explanation	Units	Unit Cost	Total Cost	Unit	Unit Cost	Total Cost	Units	Unit Cost	Total Cost
August 1	Beg. Invent.							60	$10.00	$600.00
12	Purchases	90	$12.00	$1,080.00				60	10.00	600.00
								90	12.00	1,080.00
15	Sales				60	$10.00	$600.00			
					20	$12.00	240.00	70	12.00	840.00
21	Purchases	120	13.00	1,560.00				70	12.00	840.00
								120	13.00	1,560.00
22	Sales				70	$12.00	840.00			
					30	13.00	390.00	90	13.00	1,170.00
28	Sales				65	13.00	845.00	25	13.00	$325.00
							$2,915.00			

Ending Inventory = $325 Cost of Sales = $2,915

(b)

(b). LIFO		Purchases			Sales			Balance		
Date	Explanation	Units	Unit Cost	Total Cost	Unit	Unit Cost	Total Cost	Units	Unit Cost	Total Cost
August 1	Beg. Invent.							60	$10.00	$600.00
12	Purchases	90	$12.00	$1,080.00				60	10.00	600.00
								90	12.00	1,080.00
15	Sales				80	$12.00	$960.00	60	10.00	600.00
								10	12.00	120.00
21	Purchases	120	13.00	1,560.00				60	10.00	600.00
								10	12.00	120.00
								120	13.00	1,560.00
22	Sales				100	$13.00	1300.00	60	10.00	600.00
								10	12.00	120.00
								20	13.00	260.00
28	Sales				20	13.00	260.00			
					10	12.00	120.00			
					35	10.00	350.00	25	10.00	$250.00
							$2,990.00			

Ending Inventory = $250 Cost of Sales = $2,990

(c)

(c). Moving Average		Purchases			Sales			Balance		
Date	Explanation	Units	Unit Cost	Total Cost	Unit	Unit Cost	Total Cost	Units	Unit Cost	Total Cost
August 1	Beg. Invent.							60	$10.00	$600.00
12	Purchases	90	$12.00	$1,080.00				150	11.20	1,680.00
15	Sales				80	$11.20	$896.00	70	11.20	784.00
21	Purchases	120	13.00	1,560.00				190	12.34	2,344.00
22	Sales				100	$12.34	1,234.00	90	12.34	1,110.00
28	Sales				65	12.34	801.89	25	12.34	$308.11
							$2,931.89			

Ending Inventory = $308.11 Cost of Sales = $2,931.89

(d)

(d). Specific Identification	
Ending Inventory	
5 units @ $10	$ 50.00
12 units @ $12	144.00
8 units @ $13	104.00
25 units	$ 298.00
Cost of Sales	
55 units @ $10	$ 550.00
79 units @ $12	948.00
92 units @ $13	1,196.00
226 units	$ 2,694.00

Ending Inventory = $298 Cost of Sales = $2,694